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Industrial Internship Report on

Machine Learning & Deep Learning Applications

Project 1

Smart City Traffic Pattern Forecasting using Machine Learning

Project 2

Crop & Weed Detection using YOLOv8

Prepared by

Kuldeep Kirit Prajapati

Drs. Kiran & Pallavi Patel Global University

Executive Summary
This report presents the work completed during the six-week Industrial Internship conducted by upskill Campus and The IoT Academy in collaboration with UniConverge Technologies Pvt. Ltd. (UCT). The internship provided practical exposure to Machine Learning and Deep Learning by combining theoretical learning with hands-on project development.

During the internship, I completed two industry-oriented projects. The first project, **Smart City Traffic Pattern Forecasting**, involved analyzing traffic datasets, performing data preprocessing and exploratory data analysis, engineering features, and developing predictive models using Random Forest and XGBoost. A Streamlit web application was also developed to provide real-time traffic predictions.

The second project, **Crop & Weed Detection using YOLOv8**, focused on object detection in agricultural images. The project involved preparing a YOLO-format dataset, training a YOLOv8 model, evaluating detection performance, and building a Streamlit application that identifies crops and weeds from uploaded images.

Throughout the internship, I also studied fundamental concepts of Data Science, Machine Learning, Deep Learning, Probability and Statistics through videos, e-books, quizzes, and practical exercises. This internship strengthened my technical skills in Python, Scikit-learn, XGBoost, YOLOv8, PyTorch, Streamlit, data preprocessing, model evaluation, and deployment while providing valuable experience in solving real-world AI problems.

1 Preface

This internship report presents the knowledge, practical experience, and technical skills I acquired during my six-week Industrial Internship conducted by upskill Campus and The IoT Academy in collaboration with UniConverge Technologies Pvt. Ltd. (UCT). The internship was designed to bridge the gap between academic learning and real-world industry applications by providing hands-on exposure to Machine Learning and Deep Learning.

Throughout the internship, I explored the complete machine learning development lifecycle, including data collection, preprocessing, exploratory data analysis, feature engineering, model development, evaluation, deployment, and documentation. Along with practical implementation, I strengthened my theoretical understanding through video lectures, e-books, quizzes, and guided learning modules covering Data Science, Machine Learning, Probability & Statistics, and Artificial Intelligence concepts.

As part of the internship, I successfully developed two practical projects. The first project, **Smart City Traffic Pattern Forecasting using Machine Learning**, focused on predicting traffic volume using historical traffic data. The project involved data preprocessing, feature engineering, training multiple regression models, comparing their performance, and deploying the best-performing model using Streamlit.

The second project, **Crop & Weed Detection using YOLOv8**, focused on applying Deep Learning techniques for agricultural object detection. This project involved understanding the YOLO dataset format, training a custom object detection model, evaluating its performance, and developing a Streamlit application capable of detecting crops and weeds from uploaded images.

This internship has significantly enhanced my problem-solving ability, programming skills, and understanding of practical Artificial Intelligence applications. It also improved my knowledge of Python, Scikit-learn, XGBoost, PyTorch, YOLOv8, Streamlit, Git, and GitHub while strengthening my ability to develop complete end-to-end AI solutions.

I sincerely thank **upskill Campus**, **The IoT Academy**, **UniConverge Technologies Pvt. Ltd. (UCT)**, and **Drs. Kiran & Pallavi Patel Global University** for providing me with this valuable learning opportunity. I am also grateful to the mentors and instructors whose guidance and support throughout the internship helped me strengthen both my technical knowledge and practical implementation skills.

2 Introduction

Artificial Intelligence (AI) has transformed the way modern systems solve complex real-world problems by enabling machines to learn from data and make intelligent decisions. Among its major branches, **Machine Learning (ML)** and **Deep Learning (DL)** have become essential technologies in various industries, including transportation, healthcare, agriculture, finance, and smart city development. These technologies help automate decision-making, improve prediction accuracy, and solve problems that are difficult to address using traditional programming techniques.

Machine Learning focuses on developing algorithms that learn patterns from historical data and make predictions or decisions without being explicitly programmed for every scenario. Deep Learning, a specialized branch of Machine Learning, uses artificial neural networks to automatically learn complex features from large datasets, making it highly effective for computer vision, object detection, and image classification tasks.

During this internship, I gained both theoretical knowledge and practical experience by working on two industry-oriented projects. The first project, **Smart City Traffic Pattern Forecasting**, applies machine learning techniques to predict traffic volume based on historical traffic data. The project involved data preprocessing, exploratory data analysis, feature engineering, model development, evaluation, and deployment through a Streamlit web application.

The second project, **Crop & Weed Detection using YOLOv8**, focuses on applying deep learning techniques in precision agriculture. The project uses the YOLOv8 object detection model to identify crops and weeds from agricultural field images. This solution demonstrates how artificial intelligence can assist farmers by enabling accurate and efficient weed detection, reducing manual effort, and supporting smart farming practices.

Along with practical implementation, the internship included structured learning through video lectures, e-books, and weekly assessments covering Data Science, Machine Learning, Deep Learning, Probability, and Statistics. These learning activities strengthened my understanding of AI concepts while helping me apply them effectively in real-world projects.

Overall, this internship provided valuable exposure to the complete AI project lifecycle, including data preparation, model development, evaluation, deployment, testing, documentation, and version control using GitHub.

3 2. Organization Profile

3.1 upskill Campus

upskill Campus is an online learning platform that provides industry-oriented training programs, internships, certification courses, and skill development opportunities for students and professionals. The platform focuses on bridging the gap between academic education and industry requirements by offering practical learning experiences in emerging technologies such as Artificial Intelligence, Machine Learning, Data Science, Cloud Computing, Cybersecurity, Web Development, and Internet of Things (IoT).

Through project-based learning, mentorship, and hands-on assignments, upskill Campus enables learners to apply theoretical knowledge to real-world problems and develop skills that improve their employability.

3.2 The IoT Academy

The IoT Academy is an educational platform specializing in professional training in Artificial Intelligence, Machine Learning, Data Science, Internet of Things (IoT), Cloud Computing, and Full Stack Development. The academy offers structured learning through video lectures, practical assignments, quizzes, and real-world projects designed to enhance technical knowledge and practical implementation skills.

Its industry-focused curriculum helps students build strong technical foundations while preparing them for future careers in technology.

3.3 UniConverge Technologies Pvt. Ltd. (UCT)

UniConverge Technologies Pvt. Ltd. (UCT) is a technology-driven organization focused on delivering innovative digital solutions, technical training, and industry-oriented projects. The organization collaborates with educational institutions and learning platforms to provide internship opportunities that allow students to gain practical exposure to current technologies.

During this internship, UCT provided guidance, learning resources, project assignments, and practical exposure in Machine Learning and Deep Learning through industry-relevant projects and structured assessments.

4. Internship Objectives

The primary objectives of this internship were:

- To understand the fundamentals of Data Science, Machine Learning, and Deep Learning.
- To gain practical experience in solving real-world problems using Artificial Intelligence techniques.
- To develop skills in data preprocessing, feature engineering, and exploratory data analysis.
- To build, train, and evaluate Machine Learning and Deep Learning models.
- To deploy AI models using Streamlit for interactive user applications.
- To strengthen programming skills using Python and AI libraries.
- To improve problem-solving, analytical thinking, and model evaluation skills.
- To gain experience in project documentation, GitHub version control, and professional software development practices.

5. Project 1 – Smart City Traffic Pattern Forecasting using Machine Learning

5.1 Project Overview

Traffic congestion has become one of the major challenges faced by modern cities due to increasing urbanization and vehicle population. Predicting traffic flow in advance helps traffic authorities optimize traffic signals, reduce congestion, improve road safety, and support efficient transportation planning.

The objective of this project is to develop a Machine Learning-based traffic forecasting system capable of predicting traffic volume at different city junctions using historical traffic data. The project includes data preprocessing, feature engineering, exploratory data analysis, model training, evaluation, and deployment through a Streamlit web application.

5.2 Problem Statement

Urban traffic management systems often rely on historical traffic data to understand traffic patterns. However, manually analyzing large volumes of traffic data is time-consuming and inefficient.

The objective of this project is to develop a predictive system capable of estimating traffic volume based on historical observations, enabling smarter traffic management and assisting authorities in making data-driven decisions.

5.3 Existing System

Traditional traffic monitoring systems mainly depend on historical reports, manual observation, or fixed traffic rules. These approaches have several limitations:

- Manual analysis requires significant time and effort.
- Difficult to identify complex traffic patterns.
- Limited predictive capability.
- Inefficient traffic signal optimization.
- Poor scalability with increasing traffic data

5.4 Proposed System

The proposed system uses Machine Learning algorithms to automatically learn traffic patterns from historical data and predict future traffic volume.

The system performs:

- Data preprocessing
- Exploratory Data Analysis (EDA)
- Feature Engineering
- Machine Learning model training
- Model evaluation
- Traffic prediction
- Deployment using Streamlit

This approach provides faster, more accurate, and automated traffic predictions.

5.5 Objectives

The major objectives of this project are:

- Predict traffic volume using historical traffic data.
- Improve prediction accuracy through feature engineering.
- Compare multiple regression algorithms.
- Select the best-performing model.
- Deploy the trained model using Streamlit.
- Provide an interactive user interface for real-time prediction.

5.6 Technologies Used

Technology	Purpose
Python	Programming Language
Pandas	Data manipulation
NumPy	Numerical computations
Matplotlib	Data visualization
Seaborn	Exploratory Data Analysis
Scikit-learn	Machine Learning
XGBoost	Gradient Boosting Regression
Joblib	Model serialization
Streamlit	Web application deployment

5.7 Dataset Description

The project uses a historical traffic dataset containing traffic observations collected at multiple city junctions.

Dataset Features

Feature	Description
ID	Unique record identifier
DateTime	Date and time of observation
Junction	Traffic junction identifier
Vehicles	Number of vehicles (Target Variable)

The **Vehicles** column is the target variable, while the remaining attributes are used to predict future traffic volume.

5.8 Project Workflow

Historical Traffic Dataset
↓
Data Preprocessing
↓
Exploratory Data Analysis
↓
Feature Engineering
↓
Train-Test Split
↓
Model Development
↓
Model Evaluation
↓
Model Selection
↓
Streamlit Deployment
↓
Traffic Prediction

5.9 Data Preprocessing

Before training the machine learning models, the dataset was preprocessed to improve data quality and prepare it for analysis. Data preprocessing is a crucial step that ensures the dataset is clean, consistent, and suitable for model training.

The following preprocessing steps were performed:

- Imported the dataset using the Pandas library.
- Examined the dataset structure, data types, and dimensions.
- Checked for missing values and duplicate records.
- Converted the **DateTime** column into datetime format.
- Extracted useful temporal features such as **Year**, **Month**, **Day**, **Hour**, and **Day of Week**.
- Applied One-Hot Encoding to the **Day of Week** feature.
- Removed unnecessary columns such as **ID** and the original **DateTime** column.
- Prepared the final dataset for machine learning model training.

These preprocessing steps helped improve the quality of the dataset and enabled the models to learn meaningful traffic patterns.

5.10 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) was performed to understand the characteristics of the traffic dataset before model development. Various visualizations and statistical analyses were used to identify traffic trends and relationships among features.

The following analyses were carried out:

- Dataset overview and summary statistics.
- Distribution of vehicle counts.
- Traffic analysis across different junctions.
- Time-based traffic trend analysis.
- Correlation analysis between numerical features.
- Identification of data patterns and variations.

EDA provided valuable insights into traffic behavior and helped in selecting appropriate features for model development.

5.11 Feature Engineering

Feature engineering was performed to extract meaningful information from the available data and improve model performance.

The following features were created from the **DateTime** column:

Feature	Description
Year	Observation year
Month	Observation month
Day	Observation day
Hour	Observation hour
DayOfWeek	Day of the week

5.12 Model Development

Three different regression algorithms were implemented and compared to identify the most suitable model for traffic prediction.

- **Linear Regression**

Linear Regression was used as the baseline model to establish a reference for prediction performance.

- **Random Forest Regressor**

Random Forest Regressor was implemented to capture nonlinear relationships within the traffic data. The ensemble learning approach improved prediction accuracy by combining multiple decision trees.

- **XGBoost Regressor**

XGBoost Regressor was implemented to further improve prediction performance using gradient boosting. The model efficiently handled complex feature interactions and demonstrated strong predictive capability.

The models were trained using the processed training dataset and evaluated using the testing dataset.

5.13 Model Evaluation

The regression models were evaluated using the following performance metrics:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R^2 Score

Model Performance Comparison

Model	MAE	RMSE	R^2 Score
Linear Regression	9.58	12.69	0.605
Random Forest Regressor	2.43	3.74	0.966
XGBoost Regressor	2.43	3.74	0.966

The evaluation results indicate that both ensemble models significantly outperformed the baseline Linear Regression model. Their lower error values and higher R^2 scores demonstrate a stronger ability to learn traffic patterns from historical data.

5.14 Model Deployment

After evaluating the trained models, the selected model was saved using the Joblib library and integrated into a Streamlit web application.

The deployed application allows users to:

- Select the traffic junction.

- Enter the required date and time information.
- Predict traffic volume instantly.
- Display prediction results through an interactive interface.

The Streamlit application provides a simple and user-friendly interface, enabling real-time traffic prediction without requiring knowledge of the underlying machine learning model.

6. Project 2 – Crop & Weed Detection using YOLOv8

- **6.1 Project Overview**

Agriculture plays a vital role in ensuring food security and economic growth. One of the major challenges faced by farmers is the timely identification and removal of weeds. Weeds compete with crops for nutrients, water, sunlight, and space, resulting in reduced crop productivity. Manual weed detection is time-consuming, labor-intensive, and often inaccurate, especially in large agricultural fields.

The objective of this project is to develop a Deep Learning-based object detection system capable of identifying crops and weeds from field images using the YOLOv8 model. The project demonstrates how computer vision can be applied in precision agriculture to support efficient crop management.

- **6.2 Problem Statement**

Traditional weed detection methods rely on manual inspection, which requires significant effort and may lead to inconsistent results. Detecting weeds accurately at an early stage is essential for improving crop yield and reducing unnecessary herbicide usage.

The aim of this project is to build an automated detection system that can accurately identify crops and weeds from images using a deep learning model.

- **6.3 Existing System**

Conventional weed identification methods have several limitations:

- Manual monitoring requires considerable time and labor.

- Detection accuracy depends on human expertise.
- Difficult to monitor large agricultural fields.
- Increased operational cost.
- Delayed weed identification can negatively impact crop growth.

- **6.4 Proposed System**

The proposed system utilizes the **YOLOv8 (You Only Look Once Version 8)** object detection model to identify crops and weeds in agricultural images.

The system performs the following tasks:

- Image preprocessing
- Dataset preparation
- Model training
- Object detection
- Prediction visualization
- Deployment through a Streamlit application

This automated approach provides faster and more accurate detection, assisting farmers in making informed decisions.

6.5 Objectives

The objectives of this project are:

- Detect crops and weeds automatically from field images.
- Train a YOLOv8 object detection model using labeled data.
- Improve detection accuracy through deep learning.
- Visualize detection results with bounding boxes.

- Deploy the trained model using Streamlit.
- Demonstrate the practical application of computer vision in agriculture.

6.6 Technologies Used

Technology	Purpose
Python	Programming Language
YOLOv8 (Ultralytics)	Object Detection
OpenCV	Image Processing
NumPy	Numerical Computation
Matplotlib	Visualization
Streamlit	Web Application
VS Code	Development Environment

6.7 Dataset Description

The project uses an annotated agricultural image dataset containing crop and weed images.

The dataset consists of:

- Training images
- Validation images
- Corresponding annotation files in YOLO format
- **Classes Used**

Class ID	Class Name
0	Crop

1	Weed

Each image contains labeled bounding boxes that identify the location and class of the objects.

6.8 Project Workflow

```
Agricultural Image Dataset
        |
        ▼
   Data Preparation
        |
        ▼
   Image Annotation
        |
        ▼
  YOLO Dataset Format
        |
        ▼
    Model Training
        |
        ▼
   Model Validation
        |
        ▼
    Object Detection
        |
        ▼
  Streamlit Deployment
        |
        ▼
 Crop & Weed Prediction
```

6.9 Data Preparation

Before training the YOLOv8 model, the dataset was organized according to the required directory structure.

The following steps were performed:

- Organized training and validation images.
- Verified annotation files.
- Created the data.yaml configuration file.
- Defined class names and dataset paths.
- Checked dataset consistency before training.

Proper dataset preparation ensured that the model could be trained efficiently without errors.

6.10 Model Training

The YOLOv8 object detection model was trained using the prepared dataset.

The training process included:

- Loading the pretrained YOLOv8 model.
- Configuring dataset parameters.
- Training the model for multiple epochs.
- Monitoring training and validation loss.
- Saving the best-performing model weights.

The trained model learned to recognize crops and weeds by identifying visual patterns from annotated images.

6.11 Model Prediction

After training, the best model was used to perform object detection on unseen images.

The model generated:

- Bounding boxes around detected objects.
- Class labels (Crop or Weed).
- Confidence scores for each detection.

These predictions demonstrate the model's ability to identify objects accurately in agricultural field images.

6.12 Streamlit Deployment

The trained YOLOv8 model was integrated into a Streamlit web application.

The application enables users to:

- Upload an agricultural field image.
- Process the image using the trained model.
- Detect crops and weeds automatically.
- Display the annotated output image with detected objects.

The web application provides an intuitive interface, allowing users to perform object detection without interacting directly with the underlying deep learning model.

6.13 Project Outcome

The Crop & Weed Detection system successfully demonstrated the application of deep learning in precision agriculture. The trained YOLOv8 model was able to detect crops and weeds from field images, and the Streamlit application provided an easy-to-use interface for performing predictions. This project enhanced practical knowledge of computer vision, object detection, dataset preparation, model training, and deployment.

7. Skills Learned

During this internship, I gained both technical and professional skills through project-based learning.

- **Technical Skills**
- Python Programming
- Data Preprocessing and EDA
- Machine Learning
- Deep Learning (YOLOv8)
- Streamlit Deployment
- OpenCV
- GitHub Version Control
- **Professional Skills**
- Problem Solving
- Analytical Thinking
- Project Documentation
- Time Management
- Self-Learning

8. Challenges Faced

During the internship, I encountered several challenges that helped improve my technical skills:

- Understanding data preprocessing and feature engineering.
- Preparing datasets in the correct format for YOLOv8.

- Resolving model training and prediction errors.
- Deploying machine learning models using Streamlit.
- Debugging issues related to file paths and model loading.

These challenges enhanced my debugging, research, and problem-solving abilities.

9. Conclusion

The internship provided valuable hands-on experience in Machine Learning and Deep Learning through two real-world projects: **Smart City Traffic Pattern Forecasting** and **Crop & Weed Detection using YOLOv8**. I gained practical knowledge of data preprocessing, model development, evaluation, and deployment using Streamlit. Overall, this internship strengthened my technical skills and improved my confidence in developing AI-based applications.

10. Future Scope

The developed projects can be further enhanced by:

- Using larger and more diverse datasets.
- Improving model accuracy through hyperparameter tuning.
- Deploying applications on cloud platforms.
- Adding real-time data integration.
- Extending the solutions for large-scale industrial applications.

11. References

1. Python Documentation – https://docs.python.org/
2. Scikit-learn Documentation – https://scikit-learn.org/
3. Ultralytics YOLOv8 Documentation – https://docs.ultralytics.com/
4. Streamlit Documentation – https://docs.streamlit.io/
5. Pandas Documentation – https://pandas.pydata.org/
6. NumPy Documentation – https://numpy.org/

12. Code Submission (GitHub Link)

The complete source code for both internship projects has been uploaded to GitHub for reference and evaluation.

GitHub Repo 1:

https://github.com/kuldeepin3/upSkillCampus

Projects Included:

- Smart City Traffic Pattern Forecasting
- Crop & Weed Detection using YOLO

13 Diagrams

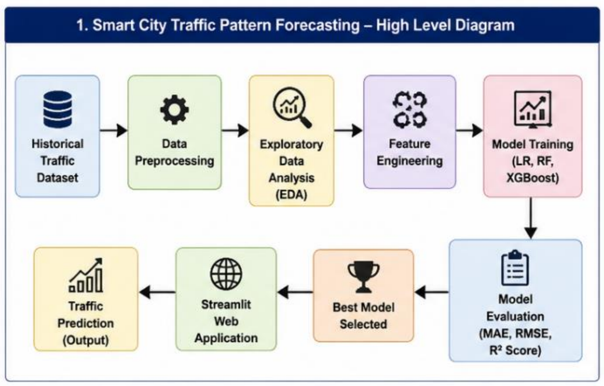

Figure 1: High Level Diagram of Smart City Traffic Pattern Forecasting

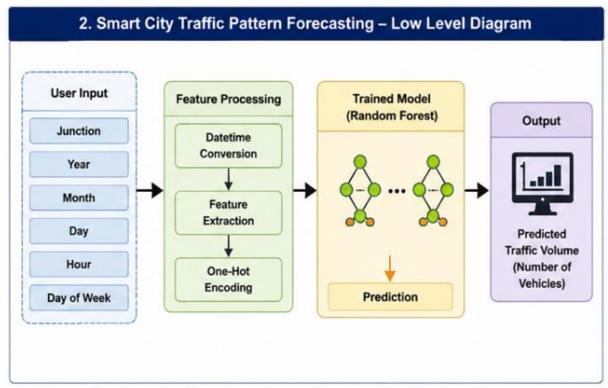

Figure 2: Low Level Diagram of Smart City Traffic Pattern Forecasting

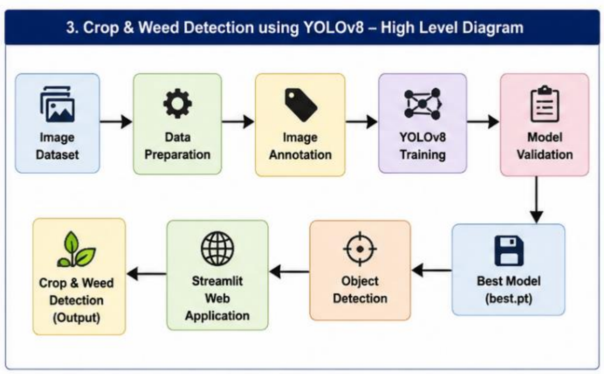

Figure 3: High Level Diagram of Crop & Weed Detection using YOLOv8

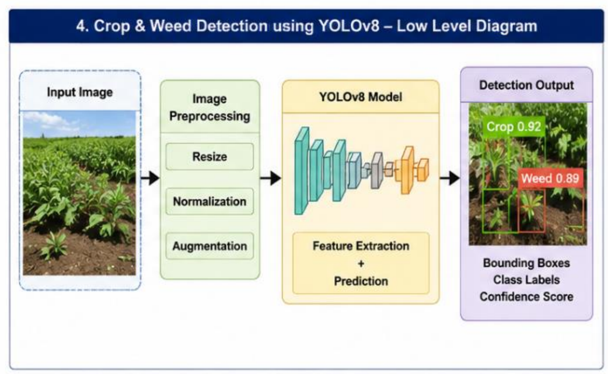

Figure 4: Low Level Diagram of Crop & Weed Detection using YOLOv8

Figure 1: High-Level Diagram of Smart City Traffic Pattern Forecasting
Paragraph:
The high-level diagram illustrates the overall workflow of the Smart City Traffic Pattern Forecasting system. The process begins with collecting the historical traffic dataset, followed by data preprocessing, exploratory data analysis (EDA), and feature engineering. Multiple machine learning models are trained and evaluated, after which the best-performing model is selected and deployed using a Streamlit web application to provide real-time traffic volume predictions.

Figure 2: Low-Level Diagram of Smart City Traffic Pattern Forecasting
Paragraph:

The low-level diagram presents the internal working of the traffic prediction system. User inputs such as Junction, Year, Month, Day, Hour, and Day of Week are processed through feature extraction and encoding before being passed to the trained Random Forest model. The model predicts the expected traffic volume, and the result is displayed through the Streamlit interface.

Figure 3: High-Level Diagram of Crop & Weed Detection using YOLOv8
Paragraph:
The high-level diagram shows the workflow of the Crop & Weed Detection system. The process starts with an annotated image dataset, followed by data preparation and YOLOv8 model training. After validation, the best-performing model is selected and integrated into a Streamlit web application, enabling users to detect crops and weeds from uploaded images.

Figure 4: Low-Level Diagram of Crop & Weed Detection using YOLOv8
Paragraph:
The low-level diagram explains the object detection process. An input image is preprocessed and passed to the trained YOLOv8 model, which extracts image features and performs object detection. The output displays bounding boxes, class labels (Crop or Weed), and confidence scores, providing accurate identification of objects in the uploaded image.